Zhen Wu

Tel: (+852) 6544 9699 E-mail: zhenwu01@cuhk.edu.hk

Summary

Postdoctoral researcher with a **Ph.D. in Geography**, specializing in **geospatial data science**, **GIS**, and **spatiotemporal modeling**, with expertise in:

- Modeling and analyzing **urban climate disasters** and **climate gentrification** under climate change.
- Modeling **carbon sinks** with **machine learning** and big data techniques, including professional **mapping** and visualization.
- Assessing **environmental and climate indices** using ***Google Earth Engine (GEE)*** and open-access satellite data.
- Conducting **spatiotemporal analyses** of **land use/land cover changes** and their socioeconomic impacts using Landsat, MODIS, and SRTM datasets.
- Optimizing **large-scale data processing and analytical workflows** through **KNIME** for efficient spatiotemporal analysis

Education

2020/09 - 2024/06 **Ph.D. in Geography**
School of Geography and Ocean Science, **Nanjing University**, Nanjing, China

2022/09 - 2023/09 **Visiting Research Scholar**
Center for Geographical Analysis (CGA), **Harvard University**, MA, US

2016/09 - 2019/06 **M.S. in Physical Geography**
School of Geographical Sciences, **East China Normal University**, Shanghai, China

2018/02 - 2018/08 **Visiting Research Scholar**
The Center for International Earth Science Information Network (CIESIN), **Columbia University**, NY, US

2011/09 - 2015/06 **B.S. in Geography Science**
School of Architecture and Urban Planning, **Hunan University of Science and Technology**, Hunan, China

Work experiences

2025/08-Present, **Postdoctoral Researcher** at The Chinese University of Hong Kong, HK SAR, China (full-time)

2024/09-2025/07, **Research Fellow** at School of Geography and Ocean Science, Nanjing University, China (full-time)

2019/07-2020/07, **Research and teaching Assistant** at Zhejiang University, China (full-time)

2025/03-2027/08, **Career Mentor** at School of Geography and Ocean Science, Nanjing University, China

2024/07-Present, **Project Mentor** at Spatial Data Lab (SDL) of Center for Geographic Analysis (CGA), Harvard University

Skills

Proficient in: Google Earth Engine (GEE) | Arc GIS Pro | Arc MAP | QGIS | KNIME | ENVI | SPSS | Origin | Python

Publications

1. **Wu, Z.,** Huang, X., Liu, X., et al. **2025.** Land use changes and their effects on poverty in Inner Mongolia, northern China. ***Geography and Sustainability***, 6(4): 100269.
2. **Wu, Z.,** Wang, X., Chen, M., et al. **2024.** Changes in sediment N mineralization and immobilization in association with Spartina alterniflora invasion in mangrove wetland. ***Plant and Soil****, 500: 495–514.*
3. **Wu, Z.,** Huang, X., Chen, R., et al. **2022.** The United States and China on the paths and policies to carbon neutrality. ***Journal of Environmental Management,*** 320: 115785. (***ESI Highly Cited Paper***)
4. **Wu, Z.,** Chen, R., Meadows, M. E., et al. **2021**. Application of the Ocean Health Index to Assess Ecosystem Health for the Coastal Areas of Shanghai, China. ***Ecological Indicators****,* 126: 107650.
5. **Wu, Z.,** Chen, R., Meadows, M. E., et al. **2019**. Changing Urban Green Spaces in Shanghai: Trends, Drivers and Policy Implications. ***Land Use Policy****,* 87: 104080. (***ESI Highly Cited Paper***)
6. **Wu, Z.,** and Chen, R. **2019**. A Comparative Analysis of OHI and PSR Evaluation Methods for Shanghai Ocean Ecosystem Health. ***Journal of East China Normal University***, 4: 174-187.

7. **Wu, Z.**, and Huang, X. **2024**. A method for estimating carbon density in terrestrial ecosystems based on multi-source remote sensing data. ***Chinese National Patent*** *(authorized),* CN118037320A.
8. Fu, X., Liu, L. Li, M., Huang, X., **Wu, Z. 2025.** Calibration of 2SFCA and i2SFCA: A Case Study in Shenzhen, China Based on Online Physician Appointment Data. ***Transactions in GIS.*** 29, e70061.
9. Huang, X., Qi, X., Liu, Z., **Wu, Z.**, et al. **2024.** Optimization of Farmland Spatial Pattern for Carbon Emission Reduction in the Agriculture and Food System. ***Economic Geography.*** 44(9), 153-162.
10. Wang, Y., Guo, C., … **Wu, Z. 2021.** Major Contribution to Carbon Neutrality by China's Geosciences and Geological Technologies. ***Journal of China Geology***, 4: 329-352.

Awards

2024 Awarded the Second Prize for Outstanding Papers from the Key Laboratory of Carbon Neutrality and Land Spatial Optimization of the Ministry of Natural Resources in China

2023 Awarded the First Prize of Excellent Scholarship from Nanjing University

2022 Awarded the Excellent Student Scholarship from Nanjing University

2017 Awarded the Third Prize winner of "The 14th China Graduate Mathematical Modelling Contest" as the team captain

Research Projects

2022/07-2025/07: ***The Project of Restoration of Degraded Ecosystems and Carbon Sink Dynamics Along the Yangtze River***

- Lead carbon sink modeling and estimation along the Yangtze River.
- Analyze spatiotemporal trends (1990–present) to identify degraded ecosystems and assess carbon sink dynamics.
- Provide policy and technical recommendations to support low-carbon development strategies.

2021/09 – 2024/06: ***The Research Project of Impacts on Climate Change on Land Use in China and Carbon Neutral Response***

- Develop carbon-related models using open-source satellite data to examine land use and carbon cycle dynamics.
- Analyze spatiotemporal changes in land use and assess carbon neutrality potential.
- Coordinate research efforts and contribute to methodological design.

2022/09-2023/09: ***Spatial Data Lab Project, Harvard University***

- Led the development of Geo-Image spatial analysis tools and KNIME workflow nodes.
- Created case studies demonstrating spatiotemporal satellite data analysis workflows.
- Organized and delivered KNIME training workshops for researchers.

2022/06-2022/12: ***The Project of Ecological Carbon Sink Potential Assessment and Carbon Emission Decrease Path in Shanghai's New City Construction***

- Co-authored project proposals and summary reports.
- Prepared technical recommendations for local government on enhancing carbon sinks and reducing emissions.

2020/12 - 2021/12: ***The Research Project of Spatial Planning in Inner Mongolia's Yellow River Basin***

- Modeled land use and ecological system dynamics under urban expansion.
- Assessed ecological vulnerability and its policy implications.
- Provided analytical reports supporting sustainable regional planning.

2017/01 - 2018/12: ***The Research Project of Ocean and Yangtze River Estuary in Shanghai's Ecological Protection***

- Processed geospatial datasets and performed environmental modeling to evaluate ecosystem health.
- Assessed coastal ecosystem changes to support sustainable development planning.

Presentations

1. Presentation "Climate-Induced Hazards and Human Impacts: A Systematic Review", ***2026 European Geosciences Union (EGU) General Assembly***, Vienna, Austria (May. 2026)
2. **Session Organizer & Chair** at the AAG conference, I organized the session "Environmental and land resource challenges in the context of climate change", and Presentation "Dynamic Evolution of Carbon Storge in China's Terrestrial Ecosystems from 1990 to 2022", ***2024 Annual Meeting of the American Association of Geographers (AAG)***, Hawaii, US (Apr. 2024)
3. Presentation "Study on the spatiotemporal evolution characteristics of carbon storage in China's terrestrial ecosystems", ***2024 World Geography Annual Conference***, Shanghai, China (Nov. 2024)
4. Presentation "Geo-AI model application in remote sensing data based on workflow", ***2024 Spatiotemporal Innovation***

Forum in Shanghai, Shanghai, China (Aug. 2024)

5. Presentation "Forest change and its carbon storage research in China from 1990 to 2021", ***The Symposium on Spatiotemporal Data Science***, Harvard University, Boston, US (Jul. 2023)
6. Presentation "Forest change and its litter carbon storage in China from 1990 to 2021", ***2023 Annual Meeting of the American Association of Geographers (AAG)***, Denver, US (Mar. 2023)
7. Presentation "A Comparative Analysis of PSR and OHI Evaluation Methods for Shanghai Ocean Ecosystem Health," ***China National Conference in Geography***, Xi'an, China (Aug. 2018)
8. Presentation "Urban Green Space Change in Shanghai", ***Global Land Program 3rd Open Science Meeting,*** China (Oct. 2016)

Internships

- **Workshop Teacher** "The Application of Remote Sensing Data in Carbon Density", *The Workshop on Spatiotemporal Innovation,* Harvard University, Boston, US (Jul. 2023)
- **Workshop Teacher** "Geo-AI models application in remote sensing data using KNIME", *The Workshop on Spatiotemporal Innovation and Data Modeling,* Harbin, China (Oct. 2023)
- **Teaching Assistant** of *Basic Introduction to GIS (Fall 2023)*, *GIS Workshop (Fall 2023), Cartography Workshop Using Arc GIS Pro (Spring 2023)*, and *Harvard GIS Institute (Winter 2022),* Harvard University, Boston, US

Referees

- Xianjin Huang, Dr. & Professor, Nanjing University, hxj369@nju.edu.cn
- Ruishan Chen, Dr. & Professor, Shanghai Jiao Tong University, rschen@sjtu.edu.cn